Midterm Exam Solution

MFE5340 - AI in FE: Quantitative Investment

Spring 2026

Question 1 (20 Points)

1. When a risk-free rate exists, the mean-variance efficient frontier becomes a straight line starting from the risk-free rate and tangential to the risky asset frontier. (True / False)

   True. With a risk-free rate, the efficient frontier changes to the capital market line (CML), which is a straight line tangent to the risky assets' frontier at the tangency portfolio.

2. The stochastic discount factor (SDF) must be positively correlated with any asset's return on the mean-variance frontier. (True / False)

   False. The SDF is inversely related to returns—when the SDF is high, returns are typically low and vice versa. This negative correlation explains risk pricing.

3. The Newey-West standard errors adjust for both heteroskedasticity and autocorrelation. (True / False)

   True. Newey-West standard errors are designed to provide robustness to both heteroskedasticity and autocorrelation. The accuracy of the Newey-West correction depends on how the lag structure is specified.

4. According to the formula in the lecture note/slides, the Elastic Net estimator reduces to Lasso when $\gamma_2 = 0$ and reduces to Ridge when $\gamma_1 = 0$. (True / False)

   True. The Elastic Net combines both L_1 and L_2 penalties: $\gamma_1 \sum |b_i| + \gamma_2 \boldsymbol{b}^\top \boldsymbol{b}$. Setting $\gamma_2 = 0$ removes the Ridge component, leaving pure Lasso. Setting $\gamma_1 = 0$ removes the Lasso component, leaving pure Ridge.

5. Risk Premium PCA (RP-PCA) incorporates both first and second moments of returns into the optimization framework. When the weighting coefficient $\gamma = 1$, RP-PCA reduces to traditional PCA. (True / False)

   False. While Risk Premium PCA (RP-PCA) does incorporate both first (mean) and second (variance) moments of returns into its optimization framework, RP-PCA reduces to traditional PCA when the weighting coefficient $\gamma = -1$, not $\gamma = 1$.

6. The probability of making at least one Type I error increases as the number of hypotheses tested increases. (True / False)

   True. Testing more hypotheses increases the chance of false positives (Type I errors).

7. The minimum Bayes factor proposed by Harvey (2017) provides the strongest deviation against the null hypothesis. (True / False)

   True. The minimum Bayes factor assumes the prior distribution of the parameter under the alternative hypothesis is centered on the maximum likelihood estimate of the data. Therefore, it provides the strongest deviation against the null hypothesis.

8. In a PCA-based factor model, the first principal component is always the factor with the highest Sharpe ratio. (True / False)

   False. The first principal component captures the most variance, but this does not necessarily correspond to the highest Sharpe ratio, which involves both return and risk considerations.

9. When using the bootstrap method for sampling factor return panel data, block bootstrap can be applied to preserve cross-sectional correlation. (True / False)

   False. Block bootstrap is indeed a method used to maintain the autocorrelation structure in time-series data by resampling blocks of observations.

10. There are no benefits from diversification if everyone has the same risk preferences. (True / False)

    False. Diversification benefits depend on return covariances, not preferences. As long as assets are imperfectly correlated, combining them in a portfolio reduces total variance below the weighted average of individual variances.

Question 2 (10 Points)

Solution:

- **(5 Points)** The risk-free rate $R_f = 3\%$, and the expected return on a portfolio with $\beta = 1.2$ is 9%. Using CAPM for the portfolio with $\beta = 1.2$:

$$9\% = 3\% + 1.2 \times (\mathbb{E}[R_m] - 3\%).$$

  Solve for $\mathbb{E}[R_m]$ and we get $\boxed{\mathbb{E}[R_m] = 8\%}$.

- **(5 Points)** First, calculate the expected return using CAPM:

$$\mathbb{E}[R] = 3\% + 2.0 \times (8\% - 3\%) = 3\% + 10\% = 13\%.$$

Now, calculate the stock's actual expected return:

$$\text{Actual expected return} = \frac{D_1 + P_1 - P_0}{P_0} = \frac{5 + 85 - 80}{80} = \frac{10}{80} = 12.5\%.$$

The actual return is less than the return required for the level of risk. That means the price is too high relative to the risk — investors are paying more than the stock is worth according to CAPM. The stock is $\boxed{\text{overpriced}}$.

Question 3 (15 Points)

Solution:

We use the Lagrange multiplier method to solve this constrained optimization problem. The Lagrangian is:

$$\mathcal{L}(\boldsymbol{w}_2, \lambda_2, \mu) = \boldsymbol{w}_2^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 - \lambda_2(\boldsymbol{w}_2^\top \boldsymbol{w}_2 - 1) - \mu \boldsymbol{w}_2^\top \boldsymbol{w}_1. \quad \text{(5 points)}$$

Taking the derivative with respect to $\boldsymbol{w}_2$:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{w}_2} = 2\boldsymbol{\Sigma}\boldsymbol{w}_2 - 2\lambda_2 \boldsymbol{w}_2 - \mu \boldsymbol{w}_1 = 0.$$

Simplifying this gives:

$$\boldsymbol{\Sigma}\boldsymbol{w}_2 = \lambda_2 \boldsymbol{w}_2 + \frac{\mu}{2}\boldsymbol{w}_1. \quad \text{(3 points)}$$

This equation implies that $\boldsymbol{w}_2$ is an eigenvector of $\boldsymbol{\Sigma}$ **as long as** $\mu = 0$. We can prove that $\mu = 0$ by multiplying both sides of this equation by $\boldsymbol{w}_1^\top$:

$$\boldsymbol{w}_1^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 = \boldsymbol{w}_1^\top \left(\lambda_2 \boldsymbol{w}_2 + \frac{\mu}{2}\boldsymbol{w}_1 \right).$$

Expanding this:

$$\boldsymbol{w}_1^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 = \lambda_2 \boldsymbol{w}_1^\top \boldsymbol{w}_2 + \frac{\mu}{2}\boldsymbol{w}_1^\top \boldsymbol{w}_1.$$

Since $\boldsymbol{w}_1^\top \boldsymbol{w}_2 = 0$ (orthogonality), the first term on the right-hand side vanishes, and this leaves:

$$\boldsymbol{w}_1^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 = \frac{\mu}{2}\boldsymbol{w}_1^\top \boldsymbol{w}_1.$$

Since $\boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1$ (normalization), we have $\boldsymbol{w}_1^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 = \frac{\mu}{2}$. From the eigenvector property of $\boldsymbol{w}_1$, i.e., $\boldsymbol{\Sigma}\boldsymbol{w}_1 = \lambda_1 \boldsymbol{w}_1$, we know that $\boldsymbol{w}_1^\top \boldsymbol{\Sigma} \boldsymbol{w}_2 = \lambda_1 \boldsymbol{w}_1^\top \boldsymbol{w}_2 = 0$. Therefore $\mu = 0$ (3 points), and $\boldsymbol{\Sigma}\boldsymbol{w}_2 = \lambda_2 \boldsymbol{w}_2$. This shows that $\boldsymbol{w}_2$ is an eigenvector of $\boldsymbol{\Sigma}$ corresponding to the second largest eigenvalue λ_2, because the objective function maximizes $\boldsymbol{w}_2^\top \boldsymbol{\Sigma} \boldsymbol{w}_2$ (4 points).

Note to grader: Students may not follow the exact procedure. As long as they show the desired conclusion with a correct and detailed derivation, full credits shall be granted to them accordingly.

Question 4 (30 Points)

Solution:

- **(6 Points)** Starting from the ridge estimator:

$$\hat{\boldsymbol{b}} = \left(\bar{\boldsymbol{\Sigma}} + \gamma \mathbf{I}\right)^{-1} \bar{\boldsymbol{\mu}}.$$

Step 1. Apply the given identity with $\bar{\boldsymbol{\Sigma}} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$:

$$\hat{\boldsymbol{b}} = \mathbf{Q}(\boldsymbol{\Lambda} + \gamma \mathbf{I})^{-1}\mathbf{Q}^\top \bar{\boldsymbol{\mu}}.$$

Step 2. Project into the PC space by premultiplying by $\mathbf{Q}^\top$ and use $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$:

$$\hat{\boldsymbol{b}}_p = \mathbf{Q}^\top \hat{\boldsymbol{b}} = \mathbf{Q}^\top\mathbf{Q}(\boldsymbol{\Lambda} + \gamma \mathbf{I})^{-1}\mathbf{Q}^\top \bar{\boldsymbol{\mu}} = (\boldsymbol{\Lambda} + \gamma \mathbf{I})^{-1}\bar{\boldsymbol{\mu}}_p.$$

Step 3. Since $(\boldsymbol{\Lambda} + \gamma \mathbf{I})$ is diagonal, the k-th element is simply:

$$\boxed{\hat{b}_{p,k} = \frac{\bar{\mu}_{p,k}}{\lambda_k + \gamma}.}$$

Grading: 2 points for correctly applying the matrix identity; 2 points for the projection step and use of orthogonality; 2 points for the final scalar expression and connection to the OLS form.

- **(6 Points)**

(i) Limiting values.

$$\lim_{\lambda_k \to \infty} \frac{\lambda_k}{\lambda_k + \gamma} = 1, \qquad \lim_{\lambda_k \to 0^+} \frac{\lambda_k}{\lambda_k + \gamma} = 0.$$

Interpretation. High-variance PCs (large λ_k) have shrinkage coefficients close to 1, so their OLS estimates are preserved almost unchanged. Low-variance PCs (small λ_k) have shrinkage coefficients close to 0, so their coefficients are shrunk aggressively toward zero. Ridge therefore applies *differential* shrinkage: stronger for low-variance PCs, weaker for high-variance PCs.

(ii) Is the student's claim correct?

No. The claim would be true only if every PC had the same eigenvalue $\lambda_k = \lambda$, in which case the shrinkage factor $\lambda/(\lambda + \gamma)$ would be a common constant. In general, λ_k varies across PCs, so each coefficient is multiplied by a *different* factor $\lambda_k/(\lambda_k + \gamma)$. This is fundamentally different from multiplying $\hat{\boldsymbol{b}}^{\text{OLS}}$ by a single scalar: ridge selectively preserves directions of high variance and suppresses directions of low variance, which a uniform scaling cannot achieve.

Grading: 1 point each limit (2 points); 1 point interpretation of differential shrinkage; 1 point correct verdict on the claim; 2 points explanation of why eigenvalue heterogeneity matters.

- **(4 Points)**

  (i) Information vs. noise. High-variance PCs capture directions in the factor space along which returns vary the most across assets and time. These directions are more likely to reflect genuine, pervasive economic risks (e.g., market-wide or sector-wide shocks) that are well identified in the data. Low-variance PCs, by contrast, capture directions of minimal return variation; with limited time-series variation to identify the corresponding SDF coefficient, these directions are dominated by estimation noise rather than economically meaningful risk premia.

  (ii) Overfitting when K is large relative to T. When K is large, OLS must estimate many coefficients from the same T observations. Low-variance PCs have small λ_k, which means the OLS estimator $\hat{b}_{p,k}^{\text{OLS}} = \bar{\mu}_{p,k}/\lambda_k$ amplifies any noise in $\bar{\mu}_{p,k}$ by dividing by a near-zero number. Applying strong shrinkage to these directions prevents the SDF from fitting idiosyncratic noise in the sample, thereby improving OOS performance. High-variance PCs are well-identified and need less correction, so light shrinkage suffices.

  Grading: 2 points for the information/noise distinction; 2 points for connecting small λ_k to noise amplification and overfitting.

- **(4 Points)**

  The student's claim is **incorrect**.

  Role of γ_2 (L_2 / ridge penalty): The term $\gamma_2 \boldsymbol{b}^\top \boldsymbol{b}$ penalizes the squared magnitude of each coefficient. It shrinks all coefficients continuously toward zero but never sets any coefficient *exactly* to zero. Increasing γ_2 produces a smoother, more uniformly small solution, but does not change the number of nonzero coefficients — the model remains dense.

  Role of γ_1 (L_1 / lasso penalty): The term $\gamma_1 \sum_i |b_i|$ penalizes the absolute magnitude. Because the L_1 ball has corners at the coordinate axes, the optimum tends to occur at a corner where some coefficients are exactly zero. Increasing γ_1 therefore induces *sparsity*: it sets more coefficients to exactly zero and reduces the number of active factors.

  Conclusion: Sparsity is controlled exclusively by γ_1. Increasing γ_2 alone cannot produce a sparser model.

  Grading: 1 point correct verdict; 1 point L_2 role (continuous shrinkage, no zeros); 1 point L_1 role (exact zeros, sparsity); 1 point identifying γ_1 as the sparsity parameter.

- **(5 Points)**

  The student's conclusion is **overstated**.

  Conditions for good OOS performance:

- ⋆ **Panel (a) — raw factors:** The out-of-sample R^2 is highest when the model is dense (i.e. the number of nonzero coefficients is large) and the ℓ_2 penalty is appropriate. If the ℓ_2 penalty is too strong (small κ), the model over-shrinks the coefficients, leading to poor performance. Similarly, if the model is too sparse (very few nonzero coefficients), important predictors are excluded, reducing R^2. This suggests that for raw factors, a dense model with appropriate amount of regularization achieves optimal out-of-sample performance.
- ⋆ **Panel (b) — PC factors:** The use of PC factors improves performance because the high-variance PCs capture most of the explanatory power, allowing the model to perform well even with fewer nonzero coefficients. Therefore, the model can be sparse in terms of PC factors. However, the ℓ_2 penalty still plays a critical role to ensure the model achieves good OOS performance.

Grading: 1 point correct verdict on the claim; 2 points describing conditions for Panel (a); 2 points describing conditions for Panel (b).

- **(5 Points)**

  While PCA has demonstrated strong empirical results when applied to raw factors, its application to individual assets, such as stocks, presents significant limitations. A major limitation is that the principal components (PCs) derived from individual asset returns may fail to capture the cross-sectional variation in expected returns, as they are optimized for variance rather than economic relevance. This can lead to a disconnect between the PCs and the underlying drivers of expected returns. To address these issues, alternative approaches such as Risk-Premium PCA (RP-PCA) have been proposed. RP-PCA incorporate both 1st and 2nd moments of stock returns when estimating the latent factors.

  Secondly, both PCA and Risk-Premium PCA do not directly incorporate predictive information contained in firm characteristics of individual stocks. To resolve this challenge, the PCA with Economic Targets framework can be employed. This approach integrates managed portfolios constructed based on firm characteristics into the PCA framework, thereby guiding the estimation process of PC factors through relevant economic information, which ultimately leads to improved out-of-sample performance.

  Grading: 2.5 points for identifying each limitation correctly.

Question 5 (25 Points + 5 Bonus Points)

Solution:

- **(6 Points)**

  Not necessarily. To answer that, we need to know if the alphas got smaller.

- ⋆ A significant $\hat{\gamma}_i$ and higher R^2 reflect *time-series* fit for a single asset; they do not imply that γ_i varies enough across assets to price the cross-section.
- ⋆ The correct test is whether the intercepts $\hat{\alpha}_i$ are jointly zero after adding SMB (e.g., a GRS test). If $\hat{\alpha}_i$ remain large and significant, the two-factor model is misspecified for the cross-section even if SMB has a positive premium.

Grading: 3 points for identifying the missing cross-sectional test; 3 points pts for explaining why time-series R^2/significance is insufficient.

- **(4 Points)**

  When two sorting variables are correlated (here, small stocks tend to have high BM and vice versa), independent sorting applies *unconditional* breakpoints for each variable. Stocks that are simultaneously extreme on both dimensions (e.g., small *and* low BM, or large *and* high BM) are rare in the data, so the corresponding corner bins receive very few observations.

  Practical problem: Thin portfolios produce *noisy* return estimates with high standard errors. Factor returns constructed from these portfolios (long minus short) are unreliable, and the estimated premia may be driven by a handful of illiquid micro-cap stocks rather than a pervasive economic effect.

  Grading: 2 points for explaining why correlation causes thin corners under independent sorting; 2 points for the practical estimation problem.

- **(5 Points)**

  The colleague's claim is **partially correct but overstated**.

  What dependent sorting does: By computing value breakpoints *within* each size quintile, the sort ensures that each of the 25 bins contains approximately equal numbers of stocks. It removes the mechanical effect of size on the value breakpoints, so the value spread is measured *conditional on* size.

  Why complete orthogonality is not achieved: Stocks within a size quintile still vary in market cap. If size and BM remain correlated within quintiles, the long and short legs of the value factor can still differ systematically in average size.

  Consequence: Because size and BM are empirically correlated, the value factor's long leg (high BM) may still systematically differ from its short leg (low BM) in market cap, so the estimated value premium partially reflects a size tilt.

  Grading: 1 point correct verdict; 2 points explaining dependent sorting; 2 points residual correlation consequence.

- **(6 Points)**

First pass (time-series, N regressions). For each asset $i = 1, \ldots, N$, run a time-series regression of R^e_{it} on the factors $(R^e_{mt}, \mathrm{SMB}_t)$ over all T months:

$$R^e_{it} = \alpha_i + \beta_i R^e_{mt} + \gamma_i \,\mathrm{SMB}_t + \varepsilon_{it}.$$

This yields estimated factor loadings $(\hat{\beta}_i, \hat{\gamma}_i)$ for each asset.

Second pass (cross-sectional, T regressions). For each month $t = 1, \ldots, T$, run a cross-sectional regression of realized excess returns R^e_{it} on the first-pass betas:

$$R^e_{it} = \lambda_{0t} + \lambda_{mt}\hat{\beta}_i + \lambda_{\mathrm{SMB},t}\hat{\gamma}_i + \eta_{it}.$$

This yields monthly risk premium estimates $\hat{\lambda}_{mt}$ and $\hat{\lambda}_{\mathrm{SMB},t}$. In total, T regressions are run, one per month.

Final estimates. The risk premium estimate is the time-series average of the monthly estimates:

$$\hat{\lambda}_{\mathrm{SMB}} = \frac{1}{T}\sum_{t=1}^{T} \hat{\lambda}_{\mathrm{SMB},t}.$$

The standard error is:

$$\widehat{\mathrm{se}}(\hat{\lambda}_{\mathrm{SMB}}) = \frac{s(\hat{\lambda}_{\mathrm{SMB},t})}{\sqrt{T}},$$

where $s(\cdot)$ denotes the time-series standard deviation of the monthly estimates.

Grading: 2 points for first pass (what is estimated, N regressions); 2 points for second pass (what is estimated, T regressions); 2 points for the final estimator and standard error formula.

- **(4 Points)**

The concern is valid when the monthly risk premium estimates $\hat{\lambda}_t$ are **serially correlated** across time. Standard Fama–MacBeth standard errors treat the $\hat{\lambda}_t$ as i.i.d. draws and compute $s(\hat{\lambda}_t)/\sqrt{T}$.

In this case, Newey–West standard errors should be used. Over 60 years of monthly data, risk premia are plausibly autocorrelated due to slowly-varying macroeconomic conditions, time-varying factor exposures, and persistent return predictability.

Grading: 2 points identifying serial correlation as the condition; 1 point naming Newey–West; 1 point connecting to the 60-year panel.

- **(Bonus 5 Points)**

The researcher's conclusion conflates *statistical significance of the risk premium* with *evidence for a risk-based explanation.*

The horse-race test. Run the second-pass cross-sectional regression including *both* the SMB

beta $\hat{\gamma}_i$ and the characteristic (log market cap, $\ln \mathrm{ME}_i$):

$$R^e_{it} = \lambda_{0t} + \lambda_{\beta,t}\hat{\gamma}_i + \lambda_{\mathrm{ME},t} \ln \mathrm{ME}_i + \eta_{it}.$$

Two possible outcomes carry distinct interpretations:

- ⋆ If $\hat{\lambda}_\beta$ remains significant and $\hat{\lambda}_{\mathrm{ME}}$ is not, the data are consistent with a risk-based story: covariance with SMB, not size per se, drives expected returns.
- ⋆ If $\hat{\lambda}_{\mathrm{ME}}$ is significant but $\hat{\lambda}_\beta$ becomes insignificant once the characteristic is controlled for, the data favor a *characteristics-based* story: market cap predicts returns independently of SMB exposure.

Economic interpretation. In any risk-based model, expected returns are determined by covariance with priced factors, not by firm characteristics per se. If two stocks share the same SMB beta but differ in market cap, a risk-based model predicts they should have the *same* expected return. Finding that the characteristic predicts returns beyond the beta therefore *rejects* the risk-based interpretation.

Alternative explanation. A non-risk-based (behavioral or mispricing) story — as in Daniel and Titman (1997) — can generate exactly this pattern: investors systematically misprice small stocks, producing a return premium that is correlated with SMB betas only because small stocks happen to load on SMB, not because SMB exposure is the priced source of risk.

Grading: 2 points for setting up the horse-race regression; 1 point for interpreting beta insignificance after controlling for characteristic; 1 point for the risk-based prediction (same beta $\Rightarrow$ same expected return); 1 point for the characteristics-based alternative.